



UNIVERSITY OF PLYMOUTH

NSBM GREEN UNIVERSITY

Faculty of Computing

PUSL2022 Introduction to IOT

Internet Of Things(IOT) Group Project

IOT-Powered Smart Thermostat

Group Name: Group BR

Group Member Details		
Student ID(Plymouth)	Name	Role
10953703	Nethmal G D D S	Lead Developer
10953574	Gunathilake V P	Hardware Engineer
10953706	Silva L D M D	Data Analyst
10953390	Karunarathna C T	UI/UX Designer
10953598	Gamage S N P	Frontend Developer
10953686	Kavindi O M	Database Manager
10953586	Kavindu Sandaruwan N W H L	Backend Developer
10953576	Gunathilaka R B P	Research & Documentation

Table of Contents

Introduction	3
Objectives	4
Background Study	5
Mapping Parameters	6
Sensor Integration through Embedded System or Edge Computer Technology	6
Embedded or Edge Computer System for Data Processing	6
Communication Infrastructure	7
Data Storage on Servers/Cloud	7
Meaningful Analysis on Long-Term Stored Data	8
Interface to Visualize the Infrastructure, Management, and Administration	8
Outcome	9
Conclusion	10

Introduction

Heating and cooling their homes accounts for almost 40% of global energy consumption, with households being a major contributor to global greenhouse gas emissions (as well). Conventional thermostats typically require inefficient use of energy through basic manual controls or inflexible programming that doesn't change based on real time. This may lead to an unnecessary increase in energy consumption, higher bills and also creating an environmental problem.

Automatically, these thermostats don't learn about a person's habits or how the home is changing due to different environmental conditions, resulting in overheated or underheated homes even when unoccupied. Our project tackles this issue through development of a smart IoT powered thermostat based on accessing real time data from advanced environmental sensors integrated to optimize energy consumption. The thermostat is able to monitor indoor temperature, humidity and occupancy until they are continuously monitored and passed to it in order to make informed automated adjustments so that user comfort is optimal while energy is wasted to the lowest level.

This novel solution corresponds to the Sustainable Development Goals adopted by the United Nations. In particular, it moves energy efficiency and promotes clean energy technologies in support of SDG 7 (Affordable and Clean Energy). Also, it is addressing SDG 12 (Responsible consumption and production) related to sustainable energy consumption practices and reducing waste. The last is that the thermostat also contributes to SDG 13 (Climate Action) by significantly reducing carbon emissions caused from residential energy use.

The proposed thermostat, thanks to the use of cutting-edge technology, not only improves user comfort but also works as a vital tool to fight against climate change and minimize the usage of energy. This system demonstrates the power of smart home tech through real time data collection and automated adjustment of behaviors, supporting users to live more sustainably and make decisions that favor both their household and our planet.

Objectives

The smart thermostat project is an innovative, IoT powered solution for automating and optimizing temperature control in homes. The main objectives include:

1. **Automate Temperature Control:** Data from environmental sensors (temperature, humidity, occupancy) will feed the thermostat, and it will automatically adjust heating and cooling systems to save users from having to play around with them manually and save energy. This objective assists in realization of SDG 7 (Affordable and Clean Energy) by making clean energy technology affordable.
2. **Enable Remote Control via Mobile Application:** Users will be able to control the thermostat remotely, see hours and days of energy usage, and get environment notifications via a mobile app. This capability enables responsible energy management, enabling SDG 12 (Responsible Protection) by enabling efficient energy use and the insights to minimize consumption.
3. **Optimize Energy Consumption:** The system is integrated with occupancy sensors, which allow the temperature to be set only when the space is unoccupied to avoid wasted energy. The provision of this feature is in line with SDG 12 commitment to the responsible use of the energy.
4. **Integrate Environmental Sensors:** It watches indoor conditions all the time, fine-tuning the comfort while using the least possible amount of energy. SDG 7 is supported through this objective by forcing the use of energy that's both efficient and tailored to real time conditions.
5. **Provide Energy Usage Insights:** The prevailing energy usage data for which will be accessible through historical data in the mobile application to increase user awareness and encourage sustainable ways. Guiding users to reduce their environmental footprint is supported by SDG 12 and SDG 13 (Climate Action).
6. **Promote Sustainability:** That will enable efficient operation of HVAC systems, lowering energy bills and greenhouse gas emissions, what it is basically about climate action.
7. **Reduce Carbon Footprint:** The project works to lower the household carbon footprint by optimizing energy use, supporting SDG13 by reducing the need for electricity powered by fossil fuels.

Overall, these objectives serve simultaneously to highlight the practical yet substantial benefits to the user from the smart thermostat, efforts to make user comfortable, provide energy savings and pose substantial impact on the environment.

Background Study

Household heating and cooling systems are among the largest consumers of energy and, as a result, they account for a great deal of global energy consumption and greenhouse gas emissions. Nearly 40 percent of the world's energy goes to buildings, and heating, ventilation and air conditioning (HVAC) systems are leading offenders, according to the International Energy Agency (IEA). These systems can result in a great deal of energy waste because of inefficiencies in the regulating of them, worsening the environmental problems and making consumers pay more for their energy.

With current smart thermostat solutions like Google Nest and Ecobee making strides toward addressing any inefficiencies with basic automation features, this is where temperature is at its most challenging. Many of the existing models, however, cannot optimize on energy consumption sufficiently. Most often, these sorts of systems do not adapt on the fly to environmental variations, for example temperature, humidity, or occupancy patterns; rather they are based upon time schedules or simple algorithms and do not change dynamically over time.

Using Internet of Things (IoT) technology, this project aims to increase the capabilities of smart thermostats. Our approach consists of a paradigm shift from being a passive user and user of a health monitoring system to an active participant in developing a network of advanced sensors that can monitor indoor conditions. Sensors on these devices will monitor values like temperature, humidity and occupancy levels before the thermostat can make intelligent real time adjustments for HVAC systems. Through automation of these processes, our smart thermostat can drastically reduce unnecessary energy consumption, saving both homeowners money and reducing the environmental impact.

The project aligns with several Sustainable Development Goals (SDGs): it helps to achieve SDG 7 (Affordable and Clean Energy), by promoting energy efficiency; encourages responsible energy consumption in line with SDG 12; and contributes to climate action by reducing greenhouse gas emissions in accordance with SDG 13. This project focuses on how real time energy usage optimization can offer a more efficient and sustainable solution to modern households in need of innovation in energy management.

Mapping Parameters

a) Sensor Integration through Embedded System or Edge Computer

Technology:

The real environmental data is collected by multiple types of sensors that are integrated into the IoT powered thermostat. Key sensors include:

- **Temperature Sensors:** They monitor temperature indoors to make sure indoor heating and cooling systems react in real time. As one example, if the summer temperature indoors exceeds a preset limit, the thermostat will bring in the air conditioning in automatic mode to return comfort.
- **Humidity Sensors:** Regions where humidity levels affect comfort will require these. For instance, the system will focus on the dehumidification during high humidity to decrease energy used for cooling without compromising the indoor suitable humidity level.
- **Occupancy Sensors:** They detect if a room is occupied. The thermostat can modify the HVAC system to use less energy if the thermostat finds no movement in one space for a set time. For instance, the system can decrease the heating or cooling if the house is empty during work times in hopes of saving energy, then will return house to the suitable temperature when people show up.

Local processing of this sensor data is achieved by using embedded systems (e.g., Raspberry Pi, Arduino, or ESP32) that can minimize the processing load and speed up decision making without relying on cloud-based processors all of the time. It eliminates latency and guarantees that if something happens to make the home uncomfortable, adjustments are made in real time to keep the home comfortable while at the same time saving energy.

b) Embedded or Edge Computer System for Data Processing:

The sensor data is processed by the embedded system in the thermostat in real time. For instance, as temperature information shows the number is 2°C more than the comfort limit, the system is being processed and it is concluded that the air conditioning has been switched on.

For example, a Raspberry Pi or an ESP32 microcontroller can also process multiple sensor inputs with programmed algorithms to control HVAC adjustments and synchronize data with cloud servers and monitor remotely. It is powerful enough to offload high level data processing tasks (e.g., analyzing energy trends) to the cloud while running locally on

the home environment to manage it. Both local decisions making in an efficient manner, and deep cloud-based analysis for long term optimization, are allowed by this.

c) Communication Infrastructure (Wi-Fi, 4G, LTE):

The intermediary component of the communication infrastructure will give the thermostat access to both the home network (through Wi-Fi) and to a wider variety of external networks (via 4G/LTE) for remote connectivity and data transfer. Here's how it works:

- **Wi-Fi Communication:** The home tracks the thermostat's readings with other smart home devices and the cloud through the home's Wi-Fi network. The system can be controlled remotely by users by means of a mobile app. As an example, an app, if the user is away from home and sees that the HVAC is running when it doesn't need to be using the app, the user can turn it off.
- **4G/LTE Connectivity:** If Wi-fi isn't provided or reliable, then the thermostat will switch over to 4 G or LTE communications to be sure of communication being uninterrupted. For example, an LTE control mechanism is still possible even when the home's Wi-Fi is down and the user isn't there, for example, to maintain control of the air environment.

In tandem, this dual communication setup guarantees that users can always monitor and control their energy consumption from a distance, improving convenience and usefulness when you are thinking of responsible energy management.

d) Data Storage on Servers/Cloud:

For long term analysis, vast amounts of real time and historical data is collected by the thermostat and stored in the cloud. This includes:

- **Temperature and Humidity Data:** This type of datasets are stored on the cloud for long term analysis of patterns in household energy consumption and indoor climate conditions. For example, users can check out a plot over a year of how their home's heating needs vary over time, allowing them to schedule more efficient HVAC cycles.
- **Energy Consumption Logs:** Energy use by day, week or month is also stored along with data from heating and cooling activity. Such information is invaluable to understand the habits of energy consumption. For instance, if energy bills increase during the summer the app can tell you how much more energy was burned for cooling and suggest new settings to help save money.

This long-term data storage enables sustainable consumption (SDG 12) by supporting energy monitoring and behavioural changes.

e) Meaningful Analysis on Long-Term Stored Data:

Passively stored data is not just stored, but actively analysed to produce actionable insights. This analysis allows for:

- **Energy Usage Forecasting:** The system is able to forecast future energy needs and costs, based on past usage patterns. Say the data indicates exactly that: energy consumption has an uptick every year in the first month of summer, the system can suggest that the cooling settings be adjusted before the surge happens, so that energy use isn't hitting its highest peak suddenly.
- **Patterns and Trend Analysis:** The stored data the thermostat analyses will tell it if the HVAC system is running unnecessarily when it is, say, empty rooms. When the app detects these inefficiencies, it can alert the user to them and suggest that the user changes settings to align with responsible energy consumption (SDG 12).
- **Environmental Insights:** It also provides long term data about how indoor temperatures respond to external weather conditions for thermostat settings to be refined. For example, during a time span of week of humid weather, the system may automatically change how to keep the air quality and temperature control, both are more comfortable and efficient.

This level of analysis allows users to be able to make intelligent decisions about the way they use their energy, and therefore forms part of a sustainable and energy efficient lifestyle.

f) Interface to Visualize the Infrastructure, Management, and Administration:

To achieve a comprehensive and intuitive experience for the user, the user interface of the thermostat is designed. It includes:

- **Mobile Application:** Main interface of users to monitor and control home environment is the app. The system includes real-time visualization of temperature, humidity, occupancy data, energy consumption reports and the opportunity to change energy consumption settings. For instance, a user can see a live graph of their home's temperature for the day and can make changes right away if it seems necessary.

- **Customizable Settings:** As an easy to customize app, the thermostat settings like temperature schedules at the time of day or room occupancy can be customized. On top of that, it has advanced settings for those who want to take their system's energy usage even further.
- **Alerts and Notifications:** Notifications will be sent to users only about big changes (e.g. when the system notices an abnormally high energy use or when it switches to energy saving modes). For instance, if a refrigerator picks up that a room is vacant for several hours, the temperature will lower itself and an alert will go off to the user of such adjustment.

The system is designed for easy use of the complex data and analysis generated, to make it easy for users to manage their home's energy consumption and sustainability practices (SDG 7, SDG 12).

Outcome

The purpose of this project is to produce a fully working prototype that is a smart thermostat which has built in features to help optimize energy consumption and be convenient for the user. Key functionalities of this innovative system include:

- **Automated Temperature Control:** With the use of real time environmental data from sensors, the thermostat will adjust heating and cooling settings automatically to give you optimal indoor comfort. It removes the necessity of adjusting manually the way that HVAC systems function with the current environmental conditions, while eliminating energy waste.
- **Remote Management via Mobile App:** Users will be able to manage their thermostat settings remotely via a user-friendly mobile application and real time indoor condition monitoring. It allows users to easily adjust the temperature settings and receive notifications about energy usage and system performance so that users can make well educated energy consumption decisions.
- **Energy Optimization Features:** The smart thermostat will calculate when to heat and cool less using occupancy sensors and smart algorithms when the spaces are unoccupied. Thanks to this proactive approach not only will users save money, but overall demand of energy will be reduced, making it more environmentally sustainable.
- **Detailed Energy Consumption Reports:** But the mobile app will give complete reports on energy consumption, and a user will be able to monitor the amount of used

energy, and which strategies used to work more efficiently. This contributes SDG 12 (Responsible consumption production) and SDG 7 (Affordable and clean energy).

Through these outcomes, the project will enable a dramatic reduction of household energy consumption and help meet SDG 13 climate action efforts. Also, that smart thermostat addresses comfort and money savers, while it acts as a critical tool to stop the warming of the planet, to make the future more sustainable for all.

Conclusion

This IoT-powered thermostat project addresses a critical challenge in modern households: Heat and cooling systems being inefficient and consuming a lot of energy. The smartest thermostat intelligently optimizes HVAC operations based on real time data from a network of sensors to achieve the right balance of energy efficiency and user comfort. This is an innovative approach which minimizes energy waste, while improving the living environment for the consumers effectively.

The project very suits the Sustainable Development Goals (SDGs), more specifically SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action). It encourages households to follow responsible energy consumption practices by means of promoting energy efficient technologies, that in turn result in reduced utility bills and small environmental footprint.

Additionally, the thermostat is one of many important devices in continuing the fight against climate change by cutting greenhouse gas emissions that stem from using too much energy. In fact, this project demonstrates the great potential of smart home technologies to address the issue of energy and climate to improve a sustainable future. Not only does its successful implementation benefit individual households, but it also serves to help realize global pursuance that seek to promote environmental sustainability and resilience at the world level.